Bounce-apart calculations

antocciphysics | TEACHER KEY | CPA

Learning target: Use momentum and energy to check an elastic collision.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 3, Day 6; 3 course days later.*
Yes: sticking defines perfectly inelastic. Total momentum is conserved; kinetic energy need not be.

02. Explain a worked example (worked)
Before +3; after -1+4=+3 kg·m/s. The carts have different masses, so momentum-arrow lengths differ from velocity-arrow lengths.

03. Look and explain (model)
A stops; B moves right at 3 m/s.

04. Build the solution (practice)
$\Delta p = -1 - 3 = -4$ kg·m/s.

05. Use the idea (reason)
No. It is inelastic; momentum conservation alone is insufficient.

06. Choose a method (mixed) *Review: Unit 2, Day 13; 12 course days later.*
Velocity tangent; acceleration inward at both positions. Constant speed does not mean constant velocity.

07. Exit check (exit)
A-1 m/s; B+4 m/s.

08. STRETCH: Apply it another way (stretch)
Initial momentum 6; heavy final 2; light final momentum 4, so 4 m/s right.

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01. Bring back an earlier idea (retrieve) *Review: Unit 3, Day 6; 3 course days later.*
A stops; B moves at +5 m/s. Equal masses exchange velocities in an ideal 1 D elastic collision.

02. Explain a worked example (worked)
Before +3; after -1+4=+3 kg·m/s. The carts have different masses, so momentum-arrow lengths differ from velocity-arrow lengths.

03. Represent and interpret (model)
A stops; B moves right at 3 m/s.

04. Solve and justify (practice)
4=-2+3 vB; vB 2 m/s. KE before 8 J; after 2+6=8 J.

05. Analyze the reasoning (reason)
No. It is inelastic; momentum conservation alone is insufficient.

06. Choose a method (mixed) *Review: Unit 2, Day 13; 12 course days later.*
Speed becomes 1/4; period becomes 64 times as large. v∝r^(-1/2), T∝r^(3/2).

07. Exit check (exit)
vf 1 m/s; KE before 5 J, after 1 J. They stick and lose 4 J kinetic energy.

08. STRETCH: Apply it another way (stretch)
Initial total 7; -4+3 v=7 gives v=11/3 m/s. Impulses -14 and +14 N·s.